Math Journal 2

Isabelle Mills

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I'm finally starting a new pdf since the old one was getting so long. Unfortunately I won't be able to have hyperlinks directly to pages in the prior pdf. But oh well. If a hyperlink goes to this page, it means you should look at a previous journal.

Journal 1 contains pages 1-790.

The first thing I want to do now that it is Spring break is finally give a formal justification to the assertion I wrote on *page 759*.

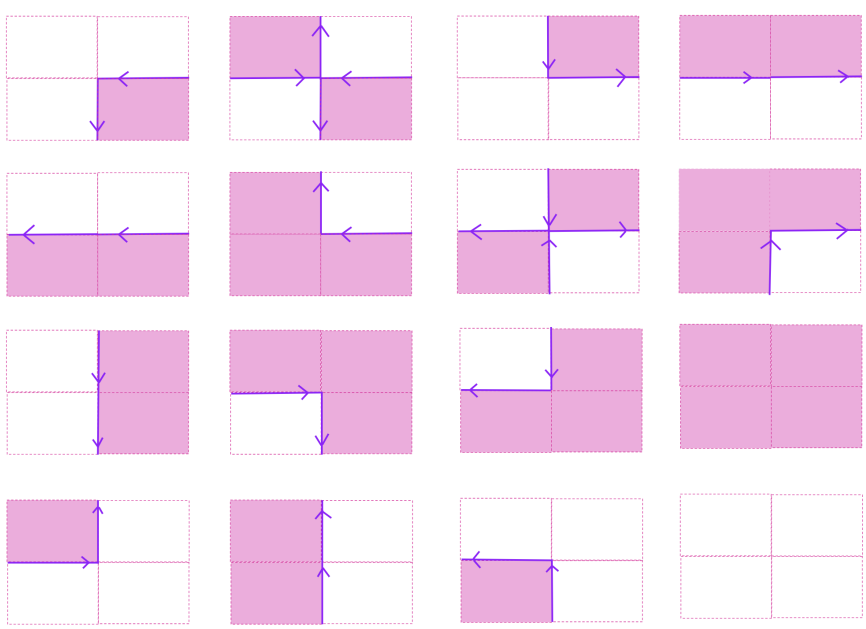
Recall the construction from *page 753*.

- Given a compact set $K \subseteq \mathbb{C}$, and $\delta > 0$, we overlay a grid of squares with side-length δ.
- Next, we consider the closed squares $Q_1, \ldots, Q_m$ that intersect K.
- Having oriented the boundary of each Q_i to be traveling counter-clockwise, we finally let $\sigma_1, \ldots, \sigma_n$ be the edges of $\partial Q_1, \ldots \partial Q_m$ which are not shared by distinct squares.

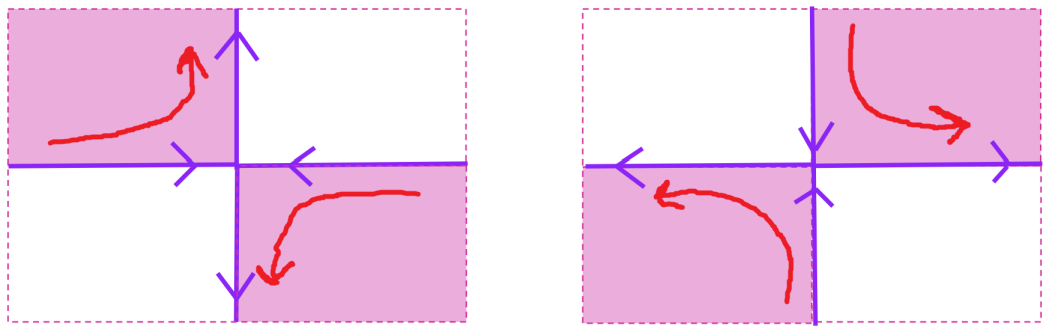
Claim: We can strategically concatenate the σ_j in order to get a bunch of continuous closed polygonal paths $\pi_1, \ldots, \pi_\ell$.

Proof:

To start off, note that an edge σ_j will move up, left, down, or right if and only if the square the edge borders to the left, below, right, or above (respectively) intersects K. By using this observation plus the fact that no σ_j are shared by two squares which both intersect K, we can thus eliminate many possible arrangements of lines. For example, here are all possible cases we can have of four adjacent squares and their shared edges:



Based on the prior image, we can assign each edge σ_j a successor edge $s(\sigma_j)$ and a predecessor edge $p(\sigma_j)$ such that $p(\sigma_j) + \sigma_j + s(\sigma_j)$ form a connected path. The only times we have any ambiguity on how we can choose $s(\sigma_j)$ and $p(\sigma_j)$ is in the two cases that we have squares diagonal to each other but not sharing any sides. Yet this isn't a big issue since for those two specific cases I can just declare the following convention.



With that, we now have that every edge has a unique predecessor and successor edge. Also, we can easily see that $p(s(\sigma_j)) = \sigma_j = s(p(\sigma_j))$. In other words, s and p are inverse functions.

Finally, consider the group action $\mathbb{Z} \curvearrowright \{\sigma_j : 1 \leq j \leq n\}$ given by $k \cdot \sigma_j = s^k(\sigma_j)$. Because there are only finitely many σ_j, we know every orbit of this group action has only finitely many elements. In particular, given any fixed σ_j we claim there exists $N \in \mathbb{N}$ such that $s^{k_1}(\sigma_j) = s^{k_2}(\sigma_j)$ iff $k_2 - k_1 \equiv 0 \pmod{N}$.

Why?

Since the orbit of σ_j is finite, we know that there exists $0 < k_1' < k_2'$ such that $s^{k_1'}(\sigma_j) = s^{k_2'}(\sigma_j)$. Then by composing on $p^{k_1} = s^{-k_1}$, we get that $\sigma_j = s^{k_2'-k_1'}(\sigma_j)$ where $k_2' - k_1' > 0$. This proves that the set of $k \in \mathbb{N}_{>0}$ such that $s^k(\sigma_j) = \sigma_j$ is nonempty. So, we can pick a minimal positive N such that $s^N(\sigma_j) = \sigma_j$. It's a basic algebra proof to go from here...

But now the orbit of σ_j consists of the collection $\{\sigma_j, s(\sigma_j), \ldots, s^{N-1}(\sigma_j)\}$. This orbit also does not intersect any distinct orbits. Also the concatenated path:

$$\sigma_j + s(\sigma_j) + \ldots + s^{N-1}(\sigma_j)$$

is connected, closed, and does not repeat any distinct edges.

Thus, we define $\pi_1, \ldots, \pi_\ell$ by considering the paths above defined from the orbits of our group action. ■

The next rabbit hole I want to descend is about $\text{weak-}L^p$. After all, this is the one topic from math 240 that even now I've not spent any time trying to learn now that the class is over and my life is somewhat solidified again.

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then recall that given any measurable $f : X \to \mathbb{C}$, we define:

- the distribution function $\lambda_f : (0, \infty) \to [0, \infty]$ of f to be:
$$\lambda_f(\alpha) := \mu(\{x \in X : |f(x)| > \alpha\});$$

- the weak-L^p quasinorm of f to be:
$$[f]_p := \left(\sup_{\alpha>0} \alpha^p \lambda_f(\alpha)\right)^{1/p} = \sup_{\alpha>0} \alpha(\lambda_f(\alpha))^{1/p} \text{ (where } 0 < p < \infty\text{)}.$$

Finally, weak-L^p (also denoted $L^{p,\infty}$) is the collection of all measurable functions f on X such that $[f]_p < \infty$.

Note that by rearranging terms, we can equivalently see that:
$$[f]_p = \min\{C \in [0, \infty] : \lambda_f(\alpha) \leq \tfrac{C^p}{\alpha^p} \text{ for all } \alpha > 0\}$$

Note that the latter set will in fact always have a minimum. The easiest way to see this is by noting that the latter set's complement (given by $\{C \geq 0 : \lambda_f(\alpha) > \frac{C^p}{\alpha^p} \text{ for some } \alpha > 0\}$) is an open set.

Thus, a function is in weak-L^p precisely iff there is some constant $D > 0$ such that $\mu(\{x : |f(x)| > \alpha\}) \leq \frac{D}{\alpha^p}$ for all $\alpha > 0$.

Unsurprisingly we have that $L^p \subseteq$ weak-L^p. After all, by Chebishev's inequality we know that $\lambda_f(\alpha) \leq \frac{\|f\|_p^p}{\alpha^p}$ for all $\alpha > 0$. Therefore, it follows that $[f]_p \leq \|f\|_p$. That said, the converse inclusion doesn't generally hold.

For example, consider the sequence $\{\frac{1}{n^{1/p}}\}_{n\in\mathbb{N}}$ as a function in $l^p(\mathbb{N})$. Its l^p norm is given by the harmonic series $\sum_{n\in\mathbb{N}} \frac{1}{n}$ which diverges. That said it is in weak-l^p. After all:
$$\lambda(\alpha) = \begin{cases} \lfloor \alpha^{-p} \rfloor & \text{if } \alpha^{-p} \notin \mathbb{N} \\ \alpha^{-p} - 1 & \text{if } \alpha^{-p} \in \mathbb{N} \end{cases}$$

Importantly, this means that $\lambda(\alpha) < \alpha^{-p}$ for all $\alpha > 0$ and we can make their difference arbitrarily small. In turn, we can show that $\sup_{\alpha>0} \alpha^p \lambda(\alpha) = 1$.

The big use case of weak-L^p is as follows. Recall that for any $0 < p < \infty$ and measurable function f on X we have that:
$$\int |f|^p \mathrm{d}\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) \mathrm{d}\alpha$$

But now if $f \in$ weak-L^p, we can say that:
$$p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) \mathrm{d}\alpha \leq p[f]_p^p \int_0^\infty \tfrac{1}{\alpha} \mathrm{d}\alpha$$

In some sense, the latter integral only just barely doesn't converge. So, given slightly more information bounding our function, we can say that f is in L^p. To further explore this, here is an exercise from Folland's *Real Analysis*.

(Folland) Exercise 6.36: If $f \in \text{weak-}L^p$ and $\mu(\{x : f(x) \neq 0\}) < \infty$, then $f \in L^q$ for all $q < p$.

Proof:

If $C := \mu(\{x : f(x) \neq 0\}) < \infty$, then we know that $\lambda_f(\alpha) \leq C$ for all $\alpha > 0$. In turn, for all $0 < q < p$ we have that:

$$\begin{aligned}\int |f|^q \mathrm{d}\mu = q\int_0^\infty \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha &= q\int_0^1 \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha + q\int_1^\infty \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha \\ &\leq qC\int_0^1 \alpha^{q-1}\mathrm{d}\alpha + q\int_1^\infty \tfrac{\alpha^{q-1}}{\alpha^p}[f]_p^p\mathrm{d}\alpha \\ &= C + [f]_p^p\tfrac{q}{p-q}.\end{aligned}$$

On the other hand, if $f \in (\text{weak-}L^p) \cap L^\infty$, then $f \in L^q$ for all $q > p$.

Proof:

If $C := \|f\|_\infty < \infty$, then we know that $\lambda_f(\alpha) = 0$ for all $\alpha \geq C$. Hence, we can say for all $p < q < \infty$ that:

$$\begin{aligned}\int |f|^q \mathrm{d}\mu = q\int_0^\infty \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha &= q\int_0^C \alpha^{q-1}\lambda_f(\alpha)\mathrm{d}\alpha \\ &\leq q\int_0^C [f]_p^p\tfrac{\alpha^{q-1}}{\alpha^p}\mathrm{d}\alpha = \tfrac{q}{q-p}C^{q-p}. \blacksquare\end{aligned}$$

One other thing on this topic I'd like to note is that $[\cdot]_p$ is in fact a quasinorm. After all:

- $[f]_p \geq 0$ with $[f]_p = 0$ if and only if $f = 0$ a.e.

  $[f]_p \geq 0$ because $\alpha^p\lambda_f(\alpha) \geq 0$ for all α. Also, because $\alpha^p > 0$ for all $\alpha > 0$, we have that $\sup_{\alpha>0}\alpha^p\lambda_f(\alpha) = 0$ if and only if $\lambda_f(\alpha) = 0$ for all $\alpha > 0$. Yet the latter will happen if and only if $\mu(\{x : f(x) \neq 0\}) = 0$.

- For any $c \in \mathbb{C}$ we have that $[cf]_p = |c|[f]_p$.

  Why?

  If $c = 0$ then this is trivial from the last bullet point. Meanwhile, if $c \neq 0$ then:

  $$\lambda_{cf}(\alpha) = \mu(\{x : |cf(x)| > \alpha\}) = \mu(\{x : |f(x)| > \tfrac{\alpha}{|c|}\}) = \lambda_f(\tfrac{\alpha}{|c|})$$

  Therefore:

  $$\sup_{\alpha>0}\alpha^p\lambda_{cf}(\alpha) = \sup_{\alpha>0}\alpha^p\lambda_f(\tfrac{\alpha}{|c|}) = |c|^p\sup_{\alpha>0}(\tfrac{\alpha}{|c|})^p\lambda_f(\tfrac{\alpha}{|c|}) = |c|^p\sup_{\beta>0}\beta^p\lambda_f(\beta)$$

- There exists $B \geq 1$ such that $[f+g]_p \leq B([f]_p + [g]_p)$ for all measurable functions f and g.

  Why?

  First note that:

  $$\begin{aligned}\lambda_{f+g}(\alpha) = \mu(\{x : |f(x)+g(x)| > \alpha\}) &\leq \mu(\{x : |f(x)| + |g(x)| > \alpha\}) \\ &\leq \mu(\{x : |f(x)| > \tfrac{\alpha}{2}\} \cup \{x : |g(x)| > \tfrac{\alpha}{2}\}) \\ &\leq \mu(\{x : |f(x)| > \tfrac{\alpha}{2}\}) + \mu(\{x : |g(x)| > \tfrac{\alpha}{2}\}) \\ &= \lambda_f(\tfrac{\alpha}{2}) + \lambda_g(\tfrac{\alpha}{2})\end{aligned}$$

  In turn $\sup_{\alpha>0}\alpha^p\lambda_{f+g}(\alpha) \leq \sup_{\alpha>0}\alpha^p(\lambda_f(\tfrac{\alpha}{2}) + \lambda_g(\tfrac{\alpha}{2})) \leq \sup_{\alpha>0}\alpha^p\lambda_f(\tfrac{\alpha}{2}) + \sup_{\beta>0}\beta^p\lambda_g(\tfrac{\beta}{2})$.

Next note that $\sup_{\alpha>0} \alpha^p \lambda_f(\frac{\alpha}{2}) = 2^p \sup_{\alpha>0} (\frac{\alpha}{2})^p \lambda_f(\frac{\alpha}{2}) = 2^p [f]_p^p$.

Similarly, we have that $\sup_{\beta>0} \beta^p \lambda_f(\frac{\beta}{2}) = 2^p [g]_p^p$. Thus, we can conclude:

$$[f+g]_p^p \leq 2^p([f]_p^p + [g]_p^p)$$

In other words, $[f+g]_p \leq 2([f]_p^p + [g]_p^p)^{1/p}$. To finish off, if $p \geq 1$ then this proves that $[f+g]_p \leq 2([f]_p + [g]_p)$.

You can justify to yourself that $(a^p + b^p)^{1/p} \leq a + b$ for all $a, b \geq 0$ when $p \geq 1$ by considering that the p-norm on $\mathbb{R}^2$ satisfies the triangle inequality.

Meanwhile, if $0 < p < 1$, then we need to find a constant $A > 1$ such that $([f]_p^p + [g]_p^p)^{1/p} \leq A([f]_p + [g]_p)$.

Tangent: While I'm finding this constant A, I shall also prove that L^p is a quasinormed space when $0 < p < 1$.

<u>Lemma 1:</u> If $a, b \geq 0$ and $0 < p < 1$, then $(a+b)^p \leq a^p + b^p$.

Proof:

If $a = 0$ then this is obvious. Meanwhile, if $a \neq 0$ then we have that $(a+b)^p \leq a^p + b^p$ if and only if $(1 + \frac{b}{a})^p = (\frac{a+b}{a})^p \leq \frac{a^p + b^p}{a^p} = 1 + (\frac{b}{a})^p$. Therefore, it suffices to show that $(1+t)^p \leq 1 + t^p$ for all $t \geq 0$.

Set $h(t) = 1 + t^p - (1+t)^p$. We want to show that $h(t) \geq 0$ for all $t \geq 0$. Fortunately, note that $h(0) = 0$. Also $h'(t) = pt^{p-1} - p(1+t)^{p-1}$. Since $p - 1 < 0$ and $0 < t < 1 + t$ implies that $(1+t)^{p-1} < t^{p-1}$, we can conclude that $h'(t) > 0$ for all $t > 0$. This proves that h is strictly increasing for $t > 0$. Hence, we can conclude that $h(t) \geq 0$ for all $t \geq 0$. ■

As a side note, mirror reasoning shows that $h(t)$ is strictly decreasing for $t > 0$ if $p > 1$. So, the opposite inequality will hold if $p > 1$.

<u>Lemma 2:</u> If $a, b \geq 0$ and $q \geq 1$, then $(a+b)^q \leq 2^{q-1}(a^q + b^q)$.

Proof:

In order to make it clear how someone might have discovered this lemma without seeing the constant 2^{q-1} on a google search, I shall specifically write down the process for finding some $A \geq 1$ such that $(a+b)^q \leq A(a^q + b^q)$. Then, it will just so happen to work out that the minimal A that works is 2^{q-1}.

Firstly, note for any $A \geq 1$ that if $a = 0$ then we always have that $(a+b)^q \leq A(a^q + b^q)$. So like before, it suffices to assume $a \neq 0$. Then because $(a+b)^q \leq A(a^q + b^q)$ if and only if $(1 + \frac{b}{a})^q \leq A(1 + (\frac{b}{a})^q)$, it suffices to show that $(1+t)^q \leq A(1 + t^q)$ for all $t \geq 0$.

Let $h(t) = \frac{A(1+t^q)}{(1+t)^q}$. We want to fix $A \geq 1$ such that $h(t) \geq 1$ for all $t \geq 0$. Fortunately, note that $h(0) = A \geq 1$. Also, we can calculate for $t \geq 0$ that:

$$h'(t) = \frac{Aqt^{q-1}(1+t)^q - Aq(1+t^q)(1+t)^{q-1}}{(1+t)^{2q}} = \frac{Aq(t^{q-1}(1+t) - (1+t^q))}{(1+t)^{q+1}} = \frac{Aq(t^{q-1} - 1)}{(1+t)^{q+1}}$$

It follows that h achieves an absolute minimum on $[0, \infty)$ at $t = 1$. So, it suffices to pick A with $h(1) = \frac{2A}{2^q} \geq 1$. The smallest A satisfying this is $A = 2^{q-1}$. ■

As a result of these two lemmas, we can now say for any measurable functions $f, g : X \to \mathbb{C}$ and $0 < p < 1$ that:

$$\|f + g\|_p \leq 2^{\frac{1}{p}-1}(\|f\|_p + \|g\|_p).$$

Proof:
As $t \mapsto t^p$ is a strictly increasing function, we can conclude from the triangle inequality that $|f(x) + g(x)|^p \leq (|f(x)| + |g(x)|)^p$ for all $x \in X$. In turn, by applying lemma 1 we get that:

$$\begin{aligned}\int |f(x) + g(x)|^p \mathrm{d}\mu &\leq \int (|f(x)| + |g(x)|)^p \mathrm{d}\mu \\ &\leq \int |f(x)|^p + |g(x)|^p \mathrm{d}\mu = \int |f|^p \mathrm{d}\mu + \int |g|^p \mathrm{d}\mu\end{aligned}$$

Next, as $\frac{1}{p} > 1$, we can conclude by lemma 2 plus the fact that $t \mapsto t^{1/p}$ is a strictly increasing function that:

$$\begin{aligned}\left(\int |f(x) + g(x)|^p \mathrm{d}\mu\right)^{1/p} &\leq \left(\int |f|^p \mathrm{d}\mu + \int |g|^p \mathrm{d}\mu\right)^{1/p} \\ &\leq 2^{\frac{1}{p}-1}\left(\left(\int |f|^p \mathrm{d}\mu\right)^{1/p} + \left(\int |g|^p \mathrm{d}\mu\right)^{1/p}\right). \blacksquare\end{aligned}$$

Going back to the problem of finding A such that $([f]_p^p + [g]_p^p)^{1/p} \leq A([f]_p + [g]_p)$, note that lemma 2 from the prior tangent tells us that $A = 2^{\frac{1}{p}-1}$ works. In turn this means that $[f + g]_p \leq 2^{1/p}([f]_p + [g]_p)$ if $0 < p < 1$.

As a side note, in general the $\text{weak-}L^p$ quasinorm will not be a true norm (i.e. it won't in general satisfy the triangle inequality). For an example of this, see my notes from math 240b.

Now it's significant that $[\cdot]_p$ is a quasinorm because any vector space $\mathcal{X}$ equipped with a quasinorm $[\cdot]$ can in turn be equipped with a topology turning that vector space into a topological vector space where convergence is equivalent to convergence in that quasinorm.

Describing this topology will require another tangent though unfortunately.

3/30/2026

I'd like to continue the rabbit-hole from yesterday by taking notes on the book *Modern Methods in Topological Vector Spaces* by Albert Wilanksy (see my *second citation*).

To start off, here are some conventions that Wilanksy uses. Given a topological space $(X, \mathcal{T})$ as well as some $x \in X$, we write $\mathcal{N}_x$ to denote the collection of neighborhoods of x. Also, we say a local base of neighborhoods of x is a collection $\mathcal{B} \subseteq \mathcal{N}_x$ such that for all $N \in \mathcal{N}_x$ there exists $E \in \mathcal{B}$ with $E \subseteq N$.

Beware: This differs from the definition of a neighborhood base that math 240b and Folland uses in that the sets in $\mathscr{B}$ are *not* required to be open. That said, it is still the case that if $\mathscr{B}$ is a local base of neighborhoods of $x \in X$ then a net $\langle x_i \rangle_{i \in I}$ converges to x if and only if x_i is eventually in each $E \in \mathscr{B}$.

While nets have the advantage of looking and acting like sequences, it will also be helpful to introduce a different generalization of the theory of convergence.

Given a set X (and denoting its power set as $\mathcal{P}(X)$), we say $\mathscr{F} \subseteq \mathcal{P}(X)$ is a <u>filter</u> if:

- $A, B \in \mathscr{F} \implies A \cap B \in \mathscr{F}$,
- $A \in \mathscr{F}$ and $A \subseteq E \implies E \in \mathscr{F}$,
- $\emptyset \notin \mathscr{F}$ and $X \in \mathscr{F}$.

Also, we say $\mathscr{B} \subseteq \mathcal{P}(X)$ is a <u>filterbase</u> if it is nonempty, doesn't contain $\emptyset$, and is directed by inclusion (i.e if $A, B \in \mathscr{B}$ then there exists $C \in \mathscr{B}$ with $C \subseteq A \cap B$).

Note that if $\{\mathscr{F}_i\}_{i \in I}$ is a family of filters, then $\bigcap_{i \in I} \mathscr{F}_i$ is also a filter. Therefore, it makes sense to talk about the smallest filter containing some generating set.

Given a filterbase $\mathscr{B}$, we claim the smallest filter containing $\mathscr{B}$ is the collection:

$$\mathscr{F} := \{A \in \mathcal{P}(X) : A \supseteq B \text{ for some } B \in \mathscr{B}\}.$$

Why?
It's clear that $\mathscr{F}$ must be contained in any filter containing $\mathscr{B}$. So, it suffices to justify why $\mathscr{F}$ is itself a filter. The only property not obvious from the definition of $\mathcal{F}$ is that if $A_1, A_2 \in \mathscr{F}$ then so is $A_1 \cap A_2 \in \mathscr{F}$. Fortunately, we can pick $B_1, B_2 \in \mathscr{B}$ with $A_1 \supseteq B_1$ and $A_2 \supseteq B_2$. Next, we can pick C in $\mathscr{B}$ with $C \subseteq B_1 \cap B_2 \subseteq A_1 \cap A_2$. So, $A_1 \cap A_2 \in \mathscr{F}$.

Given a topological space $(X, \mathcal{T})$ and some $x \in X$, notice that $\mathcal{N}_x$ is a filter. Also a local base of neighborhoods of x is merely a filterbase generating $\mathcal{N}_x$.

I mentioned before that filters give us an alternate description of convergence to that of nets. Here is what I mean:

(note for just this brief aside I'm going back to Folland...)

Given a topological space $(X, \mathcal{T})$ and a filter $\mathscr{F} \subseteq \mathcal{P}(X)$, we say $\mathscr{F}$ <u>converges</u> to x (written as $\mathscr{F} \to x$) iff $\mathcal{N}_x \subseteq \mathscr{F}$. This definition relates to the definition of convergence of nets as follows:

- Suppose $\langle x_i \rangle_{i \in I}$ is a net. Then its <u>derived filter</u> is the collection $\mathscr{F}$ of all $E \subseteq \mathcal{P}(X)$ such that x_i is eventually in E.

  (Hopefully it's obvious why $\mathscr{F}$ is a filter...)

  <u>Lemma:</u> The net $\langle x_i \rangle_{i \in I}$ converges to x if and only if its derived filter converges to x.

  This is clear from the fact that $\langle x_i \rangle_{i \in I}$ converges to x iff x_i is eventually in N for all $N \in \mathcal{N}_x$.

- Given any filter $\mathscr{F}$, we can view $\mathscr{F}$ as a set directed by (reverse) inclusion. Therefore, $\mathscr{F}$ can be used to index a net. We say a net $\langle x_F \rangle_{F \in \mathscr{F}}$ is associated with $\mathscr{F}$ iff $x_F \in F$ for all $F \in \mathscr{F}$.

  Lemma: A filter $\mathscr{F}$ converges to x if and only if all its associated nets converge to x.

  $(\Longrightarrow)$
  Suppose $\langle x_F \rangle_{F \in \mathscr{F}}$ is any net associated with $\mathscr{F}$. To prove that $x_F \to x$, it suffices to show that if $N \in \mathcal{N}_x$ then x_F is eventually in N. Fortunately, as $\mathscr{F} \to x$, we know that $N \in \mathscr{F}$. Then for all $F \subseteq N$ we have that $x_F \in N$.

  $(\Longleftarrow)$
  If $\mathscr{F} \not\to x$ then there must exist some neighborhood N of x such that $N \notin \mathscr{F}$. In turn, given any $F \in \mathscr{F}$ we must have that $F - N \neq \emptyset$. After all we'd have a contradiction if $F \subseteq N$.

  For each $F \in \mathscr{F}$ pick $x \in F$ with $x \notin N$. Then $\langle x_F \rangle_{F \in \mathscr{F}}$ is a net associated with $\mathscr{F}$ but it does not converge to x since x_F is not eventually in N. ■

One more definition is as follows. If V is a vector space, we say a collection $\mathscr{B} \subseteq \mathcal{P}(V)$ is additive if for each $B \in \mathscr{B}$ there exists $A \in \mathscr{B}$ with $A + A \subseteq B$.

Theorem 4-3-5: Let $\mathcal{X}$ be a real or complex vector space. Let $\mathscr{B}$ be an additive filterbase of balanced absorbing subsets of $\mathcal{X}$. Then there is a unique topology turning $\mathcal{X}$ into a topological vector space and satisfying that $\mathscr{B}$ is a local base of neighborhoods of 0.

(See *pages 230-232* for a refresher on the definitions of a balanced set and an absorbing set in a real or complex vector space. Importantly, even though I've priorly only used those terms when referring to vector spaces already equipped with some topology, the definitions of those terms are purely algebraic. So it still makes sense to talk about balanced sets and absorbing sets in any real or complex vector space.)

Proof (Existence):
We define our topology how one would reasonably expect. We shall say $U \subseteq \mathcal{X}$ is open if and only if for all $x \in U$ there exists $B \in \mathscr{B}$ with $x + B \subseteq U$.

Claim 1: This defines a topology on $\mathcal{X}$.

- It's trivial that $\emptyset$ and $\mathcal{X}$ are both open.
- Suppose U_1, U_2 are both open. Then for any $x \in U_1 \cap U_2$, we can pick $B_1, B_2 \in \mathscr{B}$ with $x + B_1 \subseteq U_1$ and $x + B_2 \subseteq U_2$. Since $\mathscr{B}$ is a filterbase, we know there exists $B \in \mathscr{B}$ with $B \subseteq B_1 \cap B_2$. In turn, $x + B \subseteq (x + B_1) \cap (x + B_2) \subseteq U_1 \cap U_2$. As x was arbitrary, this proves that $U_1 \cap U_2$ is open.

  By induction we can conclude that finite intersections of open sets are open.

- Suppose $\{U_\alpha\}_{\alpha\in A}$ is a family of open sets. Then if $x \in U := \bigcup_{\alpha\in A} U_\alpha$, we know that $x \in U_{\alpha_0}$ for some α_0. So, there exists $B \in \mathscr{B}$ with $x + B \subseteq U_{\alpha_0} \subseteq U$. Since x was arbitrary, this proves that $\bigcup_{\alpha\in A} U_\alpha$ is open.

<u>Claim 2:</u> $\mathscr{B}$ is a local base of neighborhoods of 0.

To start off, if N is a neighborhood of 0, then there exists an open set U such that $0 \in U \subseteq N$. In turn, by how we defined our topology we know there exists $B \in \mathscr{B}$ with $B = 0 + B \subseteq N$. This shows that $\mathscr{N}_0$ is contained in the filter generated by $\mathscr{B}$.

To finish showing that $\mathscr{N}_0$ is the filter generated by $\mathscr{B}$, we need to show that $\mathscr{B} \subseteq \mathscr{N}_0$ (i.e. all $B \in \mathscr{B}$ are neighborhoods of 0 in $\mathcal{X}$). So consider any $B \in \mathscr{B}$ and define $U := \{x \in \mathcal{X} : x + A \subseteq B \text{ for some } A \in \mathscr{B}\}$.

- Note that as each $A \in \mathscr{B}$ contains 0 (since each A is balanced), we know that $U \subseteq B$.
- Also $0 \in U$ as $0 + B = B$.
- Finally, consider any $x \in U$. Then by the definition of U there exists $A \in \mathscr{B}$ with $x + A \subseteq B$. And since $\mathscr{B}$ is additive, we can find a set $C \in \mathscr{B}$ with $C + C \subseteq A$. We claim that $x + C \subseteq U$. After all, suppose $y \in x + C$. Then $y + C \subseteq x + C + C \subseteq x + A \subseteq B$. This proves that $y \in U$, and since y was arbitrary, we conclude that $x + C \subseteq U$. Finally, by the definition of an open set plus the arbitrariness of x, this proves that U is open.

As an important side note, by slightly modifying the proof of the prior claim we can also show $x + B$ is a neighborhood of x for all $B \in \mathscr{B}$. Yet also since for any open set U containing x we have that there exists some $B \in \mathscr{B}$ with $x + B \subseteq U$, we can thus conclude that $x + \mathscr{B} := \{x + B : B \in \mathscr{B}\}$ is a local base of neighborhoods of x for any $x \in \mathcal{X}$. This will be used in the proof of the third and final claim.

<u>Claim 3:</u> The vector space operations are continuous with respect to our defined topology.

First we show vector addition is continuous. Consider any $x, y \in \mathcal{X}$ and let N be any neighborhood of $x + y$. Then it suffices to show that there are neighborhoods N_x and N_y of x and y respectively such that $x' + y' \in N$ for any $x' \in N_x$ and $y' \in N_y$. Fortunately, we know there exists $B \in \mathscr{B}$ with $x + y + B \subseteq N$. Next as $\mathscr{B}$ is additive, we know there exists $A \in \mathscr{B}$ with $A + A \subseteq B$. So, set $N_x = x + A$ and $N_y = y + A$. Then $N_x + N_y = x + y + A + A \subseteq x + y + B \subseteq N$.

Next, we show scalar multiplication is continuous. Consider any $c \in \mathbb{C}$ (or $\mathbb{R}$) and $x \in \mathcal{X}$. Then again let N be any neighborhood of cx. Then it suffices to find a neighborhood N_x of x in $\mathcal{X}$ and a disk Δ_c about c in $\mathbb{C}$ (or $\mathbb{R}$) such that $c'x' \in N$ for all $c' \in \Delta_c$ and $x' \in N_x$.

Let $B \in \mathscr{B}$ satisfy that $cx + B \subseteq N$. Then by the additivity of $\mathscr{B}$ we can pick $A \in \mathscr{B}$ such that $A + A + A + A \subseteq B$. Also, we can pick $D \in \mathscr{B}$ with $D \subseteq A$ and $cD \subseteq A$.

To see why, let $n \in \mathbb{N}_{>0}$ satisfy that $2^n \geq |c|$. Then choose D such that $\underbrace{D + \ldots + D}_{2^n \text{ terms}} \subseteq A$. Since D is balanced, we have that:

$$cD \subseteq 2^n D \subseteq \underbrace{D + \ldots + D}_{2^n \text{ terms}} \subseteq A$$

Since D is absorbing and contains 0, we can find some $r > 0$ such that $\alpha x \in D$ when $|\alpha| \leq r$. Let Δ_c be the disk of radius r centered at $c \in \mathbb{C}$ (or $\mathbb{R}$). Also let $N_x = x + D$. Then we claim that $\Delta_c N_x \subseteq N$.

Suppose $y \in N_x = x + D$ and $c' = c + \alpha \in \Delta_c$ where $|\alpha| < r$. Then:

$$c'y = (c+\alpha)(y - x + x) = c(y-x) + cx + \alpha(y-x) + \alpha x$$

But $y - x \in D$. Therefore, we can conclude that $c(y-x) \in cD \subseteq A$. Also as $|\alpha| < r$, we have that $\alpha x \in D \subseteq A$.

Finally, without loss of generality we can assume $r < 1$. Then as D is balanced and $|\alpha| < r$, we have that $\alpha(y-x) \in \alpha D \subseteq D \subseteq A$. So, we can conclude that $c'y \in cx + A + A + A \subseteq cx + B \subseteq N$.

Proof (Uniqueness):
To see that the topology we just defined is unique, first note that any topology satisfying our theorem must have the same collection $\mathcal{N}_0$ of neighborhoods of 0. In other words, $\mathscr{B}$ uniquely determines the neighborhoods of 0. Secondly, note that if our topology is specifically a vector space topology, then the convergence of any net is uniquely determined based on the neighborhoods of 0. Therefore, every topology satisfying our theorem will have the same nets converging. This proves that every topology satisfying our theorem is equal (see *page 229*). ■

As a special case of the prior theorem, we can prove that every quasinormed vector space (with quasinorm $[\cdot]$) has a unique vector space topology such that $\langle x_i \rangle_{i \in I}$ converges to x if and only if $[x_i - x] \to 0$.

Proof:
Let $\mathscr{B} := \{B_r = \{x \in \mathcal{X} : [x] < r\} : r > 0\}$. Checking that $\mathscr{B}$ is a filterbase consisting of balanced absorbing sets is trivial. The only nonobvious statement is that $\mathscr{B}$ is additive. Consider any $r > 0$ and pick $C \geq 1$ satisfying that $[x+y] \leq C([x]+[y])$ for all $x, y \in \mathcal{X}$. Then given any $\ell < \frac{r}{2C}$ we have for all $x, y \in B_\ell$ that $[x+y] \leq C([x]+[y]) \leq 2C\ell < r$. Therefore $B_\ell + B_\ell \subseteq B_r$.

By the prior theorem, there is thus a unique vector space topology with $\mathscr{B}$ as a local base of neighborhoods of 0. Finally, note that a net $\langle x_i \rangle_{i \in I}$ in this topology converges to 0 iff x_i is eventually in B_r for every $B_r \in \mathscr{B}$, and in turn that happens iff $[x_i] \to 0$. By applying a translation, we can thus see that the topology we constructed here is the topology of convergence with respect to this quasinorm.

As a side note, one particularly cursed observation we can make is that we shouldn't generally expect the balls $B_r := \{x : [x] < r\}$ to be open (even though they are neighborhoods of 0).

After all suppose $[x] < r$ and $[z + x] \leq C([z] + [x])$ for all $z \in \mathcal{X}$. Then if B_r were open we must be able to find a ball $B_\ell(x) := \{y : [y - x] < \ell\}$ such that $B_\ell(x) \subseteq B_r$. In other words, given that $[y - x] < \ell$ we'd want to know that

$[y] \leq C([y-x]+[x]) < r$. Yet assuming $C > 1$, we can pick x with $C[x] > r$. Therefore we'd need to have that $C\ell < r - C[x] < 0$. Yet as C and ℓ are positive, this is impossible.

On an optimistic note, you can interpret the prior reasoning as proving that $B_{r/C}$ is contained in the interior of B_r.

If we can get a sharper estimate for $[x+y]$ though, then it may be the case that all the balls are open. For example, the quasinorm topology on $\text{weak-}L^p$ does contain all the balls $\{f \in \text{weak-}L^p : [f-g]_p < r\}$ (where $g \in \text{weak-}L^p$ and $r > 0$).

Claim: Suppose $g \in B_r := \{f \in \text{weak-}L^p : [f]_p < r\}$. Then there exists a ball $B_\ell(g) := \{f \in \text{weak-}L^p : [f-g] < \ell\}$ which is contained in B_r. Hence, B_r is open.

Proof:

Given any $t \in (0,1)$, we can conclude that $\lambda_f(\alpha) \leq \lambda_{f-g}((1-t)\alpha) + \lambda_g(t\alpha)$. Then in turn, like on *pages 794-795* we have that:

$$[f]_p \leq \left(\tfrac{1}{(1-t)^p}[f-g]_p^p + \tfrac{1}{t^p}[g]_p^p\right)^{1/p}$$

Now we want to find some ℓ such that if $[f-g]_p < \ell$ then $[f]_p < r$. Fortunately, for all $t \in (0,1)$ we have that:

$$\left(\tfrac{1}{(1-t)^p}[f-g]_p^p + \tfrac{1}{t^p}[g]_p^p\right)^{1/p} < r \text{ if and only if } [f-g]_p^p < (1-t)^p(r^p - \tfrac{1}{t^p}[g]_p^p)$$

As $[g]_p < r$, we can fix t close enough to 1 such that $r^p - \frac{1}{t^p}[g]_p^p$ is positive. Then as $(1-t)^p$ is also positive, we can conclude that $(1-t)^p(r^p - \frac{1}{t^p}[g]_p^p) > 0$. So, there exists some $\ell > 0$ such that if $\ell^p < (1-t)^p(r^p - \frac{1}{t^p}[g]_p^p)$. Then $[f-g]_p < \ell$ implies that $[f]_p < r$. ■

Corollary: The quasinorm topology on $\text{weak-}L^p$ is the topology generated (in the typical topological sense) by the balls $B_\ell(g) := \{f \in \text{weak-}L^p : [f-g] < r\}$ (where $g \in \text{weak-}L^p$ and $r > 0$).

3/31/2026

Math 220C Lecture Notes:

A convention I will be using during spring quarter is that if $\Omega \subseteq \mathbb{R}^m$ is an open set and $u : \Omega \to \mathbb{R}$ is a function, then I may denote the jth partial derivative of u as u_{x_j}. Also u_{x_j,x_k} will refer to $(u_{x_j})_{x_k}$ as opposed to $(u_{x_k})_{x_j}$. Usually this distinction won't matter though since we'll typically assume we are working with functions that are C^2 if we need second order partial derivatives.

In specifically math 220c, we'll often identify $\mathbb{C}$ with $\mathbb{R}^2$ via the isometry $x+iy \mapsto (x,y)$. Thus if f is a complex variable function, we shall refer to $\partial_x f = f_x$ as the resulting partial derivative in the direction of the real axis. And similarly, $\partial_y f = f_y$ will be the partial derivative of f in the direction of the imaginary axis.

Recall from *page 332* that if $G \subseteq \mathbb{C}$ is a region, we say $u : G \to \mathbb{R}$ is a harmonic function if $u \in C^2(G)$ and $u_{x,x} + u_{y,y} = 0$.

For example, if $f \in O(G)$ then $u = \mathrm{Re}(f)$ and $v = \mathrm{Im}(f) = \mathrm{Re}(-if)$ are both harmonic functions. In this case, we say u and v are harmonic conjugates.

Lemma: If G is simply connected, then any harmonic $u : G \to \mathbb{R}$ admits a harmonic conjugate.

Proof:
Let $F = u_x - iu_y$ be a function from G to $\mathbb{C}$. Then note that $F \in C^1(G)$ and satisfies the Cauchy-Riemann equations. After all $(u_x)_x = (-u_y)_y$ by the Laplace equation (i.e. the PDE defining harmonic functions). Meanwhile $(u_x)_y = -(-u_y)_x$ is true since $u \in C^2(G)$. It follows that F is a holomorphic function. And since G is simply connected, we in turn know that $F = f'$ for some $f \in O(G)$.

Write $f(x+iy) = \alpha(x+iy) + i\beta(x+iy)$ where α, β are real-valued functions. Since f' exists (and equals F), we can get f' by differentiating in whatever direction we find convenient. In other words, we know that $u_x - iu_y = F = f' = \alpha_x + i\beta_x$. In turn, we know that $u_x = \alpha_x$ and $-u_y = \beta_x$. Since f is holomorphic and thus satisfies the Cauchy-Riemann equations, we also know $\beta_x = -\alpha_y$. So, this proves that $u_y = \alpha_y$.

Since u and α have the same partial derivatives, we can thus conclude that $u = \alpha + C$ where C is some constant. Yet without loss of generality we can assume $C = 0$. After all, if not then $f - C$ is as valid a primitive of F as f. Finally, we now have that a harmonic conjugate of u is given by the function β. ■

We can also prove the converse statement (as is shown by the following homework exercises). The key observation to make is just that even if an open set $U \subseteq \mathbb{C}$ isn't simply connected, every point in U still has a simply connected neighborhood contained in U (namely some disk of small enough radius). Therefore, any harmonic function *locally* admits a harmonic conjugate. And since satisfying a PDE is a local property, that is often enough for our purposes.

Set 1 Problem 1: If U, V are connected open sets in $\mathbb{C}$, $g : V \to U$ is holomorphic, and $u : U \to \mathbb{R}$ is harmonic, then $u \circ g : V \to \mathbb{R}$ is harmonic in V.

Proof:
Given any $z \in V$, by the continuity of g we can find disks $\Delta_z \subseteq V$ and $\Delta_{g(z)} \subseteq U$ centered at z and $g(z)$ respectively such that $g(\Delta_z) \subseteq \Delta_{g(z)}$. Then as $\Delta_{g(z)}$ is simply connected, we know u has a harmonic conjugate on $\Delta_{g(z)}$. So, we can consider a holomorphic function $f : \Delta_{g(z)} \to \mathbb{C}$ satisfying that $\mathrm{Re}(f) = u$.

To finish off, by chain rule we know that $f \circ g$ is a holomorphic function on Δ_z. Also, we have that $\mathrm{Re}(f \circ g) = u \circ g$. This proves that $u \circ g$ is a harmonic function on Δ_z. Since z was arbitrary, this proves that $(u \circ g)_{x,x} + (u \circ g)_{y,y} = 0$ everywhere on U. So, $u \circ g$ is harmonic in V. ■

Set 1 Problem 3: Let $G \subseteq \mathbb{C}$ be open and connected.

(i) Show that if $h : G \to \mathbb{C}$ is holomorphic and nowhere zero in G, then $u(z) = \log|h(z)|$ is harmonic in G.

Let $u(z) := \log|z|$. If we can show that u is harmonic in $\mathbb{C} - \{0\}$, then the statement we are trying to prove is just an application of problem 1.

Fortunately, note that $u(x+iy) = \log(\sqrt{x^2+y^2}) = \frac{1}{2}\log(x^2+y^2)$. Therefore, we can calculate that $u_x(x+iy) = \frac{x}{x^2+y^2}$ and in turn:

$$u_{x,x}(x+iy) = \frac{x^2+y^2-x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

By symmetry, we thus also have that $u_{y,y}(x+iy) = \frac{x^2-y^2}{(x^2+y^2)^2}$. Hence, $u_{x,x} + u_{y,y} = 0$.

Also $u \in C^\infty(\mathbb{C} - \{0\})$ as $t \mapsto \log(t)$ is a smooth function from $(0,\infty)$ to $\mathbb{R}$ and $(x,y) \mapsto \sqrt{x^2+y^2}$ is a smooth function from $\mathbb{R}^2 - \{(0,0)\}$ to $(0,\infty)$. This finishes the proof that u is harmonic in $\mathbb{C} - \{0\}$.

(ii) Show that G is simply connected if every harmonic function in G admits a harmonic conjugate.

Let $h \in O(G)$ be any nowhere zero holomorphic function on G. To prove that G is simply connected, it suffices to proves that G is a logarithm domain, and to do that it suffices to prove that there exists a function $g \in O(G)$ with $h(z) = e^{g(z)}$.

Yet since every harmonic function in G admits a harmonic conjugate and we know from part (i) that $\log|h(z)|$ is harmonic in G, we know that there is a holomorphic function $g \in O(G)$ with $\mathrm{Re}(g(z)0) = \log|h(z)|$. Next, we consider the function $f(z) := \frac{h(z)}{e^{g(z)}}$. Importantly, $f \in O(G)$ and $|f(z)| = \frac{|h(z)|}{e^{\mathrm{Re}(g(z))}} = \frac{|h(z)|}{e^{\log|h(z)|}} = 1$ for all $z \in G$. By the maximum modulus principle, we can thus conclude that f is a constant function (with magnitude 1). And this proves that there exists some $C \in \mathbb{C}$ with $|C| = 1$ and $Ce^{g(z)} = h(z)$. Finally, write $C = e^{is}$ for some $s \in [-\pi, \pi)$. Then $h(z) = e^{g(z)+is}$. This proves that h has a logarithm.

(iii) Show that $u(z) = \log|z|$ admits no harmonic conjugate in $G = \mathbb{C} - \{0\}$.

Suppose for the sake of contradiction that u has a harmonic conjugate $v : \mathbb{C} - \{0\} \to \mathbb{R}$. Then, we'd know the function $f(z) := u(z) + iv(z)$ is a holomorphic function in $O(\mathbb{C} - \{0\})$ satisfying that $|e^{f(z)}| = e^{u(z)} = |z|$ for all $z \in \mathbb{C} - \{0\}$. Thus by identical reasoning to that of part (ii), we can conclude that there exists some $C \in \mathbb{C}$ with $|C| = 1$ and $Ce^{f(z)} = z$ on $\mathbb{C} - \{0\}$. In turn, this would imply the existence of a branch of the logarithm function on $G = \mathbb{C} - \{0\}$.

Yet we know no branch of the logarithm exists on G.

If such a branch did exist, then we'd have to have that $\frac{1}{z}$ has a primitive on $\mathbb{C} - \{0\}$. In turn, we'd have to have that $\int_{|z|=1} \frac{1}{z}\mathrm{d}z = 0$. Yet we can calculate that $\int_{|z|=1} \frac{1}{z}\mathrm{d}z = \pm 2\pi i \neq 0$. ■

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